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May 23, 1962

FEDERAL SPECIFICATION

INSULATION BLOCK AND PIPE COVERING THERMAL
(CALCIUM SILICATE, FOR TEMPERATURES UP TO 1200°F.)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers calcium silicate in forms for use as thermal insulation.

1.2 Classification.

1.2.1 *Types and classes.* Insulation shall be furnished in the following types and classes, as specified (see 6.2):

Type:
I—Block.
II—Pipe insulation.

Class:
A—Uncovered.
B—Canvas jacketed.
C—Aluminum jacketed.
D—Kraft paper, aluminum foil jacketed.
E—Dual temperature jacketed.
F—Other jackets.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585—Boxes, Wood, Wirebound.

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-640—Boxes, Fiberboard, Corrugated, Triple-Wall.

PPP-B-650—Box, Fiberboard Special Purpose (Records Retiring).

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547—Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

MIL-M-46032—Mildew-Resistant Treatment for Fabrics, Copper Processes.

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-147—Palletized Unit Loads 40 Inch by 48 Inch, 4-Way Partial and 4-Way Pallets.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Publications:

C 165—Test for Compressive Strength of Preformed Block-Type Thermal Insulation.

C 177—Test for Thermal Conductivity of Materials by Means of the Guarded Hot Plate.

C 203—Test for Breaking Load and Calculated Flexural Strength of Preformed Block-Type Thermal Insulation.

C 302—Test for Density of Preformed Pipe-Covering Type Thermal Insulation.

C 303—Test for Density of Preformed Block-Type Thermal Insulation.

C 335—Test for Thermal Conductivity of Pipe Insulation.

C 356—Test for Linear Shrinkage of Preformed High Temperature Thermal Insulation Subjected to Soaking Heat.

C 390—Sampling Preformed Thermal Insulation.

C 446—Test for Breaking Strength and Calculated Modulus of Rupture of Preformed Insulation for Pipes.

C 533—Specification for Calcium Silicate Block and Pipe, Thermal Insulation.

D 781—Test for Puncture and Stiffness of Paperboard, Corrugated and Solid Fiberboard.

E 84—Test for Surface Burning Characteristics of Building Materials.

E 96—Test for Water Vapor Transmission of Materials in Sheet Form.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

3. REQUIREMENTS

3.1 Description.

3.1.1 *Blocks.* Blocks shall be flat, rectangular sections, or curved, rectangular sections.

3.1.2 *Pipe insulation.* Pipe insulation shall be hollow cylinders split in half lengthwise or curved segments of hollow cylinders.

3.2 Type I. Blocks shall conform to the requirements of ASTM C 533 and shall be of the shape and size, as specified (see 6.2).

3.3 Type II. Pipe insulation shall conform to the requirements of ASTM C 533. Size and thickness of hollow cylinders of pipe insulation shall conform to table I, as specified (see 6.2).

3.4 Classes.

3.4.1 *Class A.* Uncovered pipe insulation shall be furnished without jackets.

3.4.2 *Class B.* Canvas jacketed pipe insulation shall have a cotton cloth jacket securely attached to the cylinder in a manner which provides a minimum overlap of 1 inch on one end and along the cylinder seam. When specified (see 6.2), the cotton jacket shall be rendered fungus-resistant in accordance with MIL-M-40032.

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TABLE I. *Thickness of hollow cylinder pipe insulation*

Pipe size		Nominal thickness								
		1 inch	1-1/2 inches	2 inches	2-1/2 inches[1]	3 inches[1]	3-1/2 inches[1]	4 inches[1]	4-1/2 inches[1]	5 inches[1]
Nominal	Outer diameter	Actual thickness	Actual thickness	Actual thickness	Actual thickness	Actual thickness	Actual thickness	Actual thickness	Actual thickness	Actual thickness
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1/2	0.840	1	1-9/16	2-1/16	2-7/8	3-3/8	3-7/8	4-3/8	4-15/16	5-7/16
3/4	1.050	7/8	1-7/16	1-15/16	2-3/4	3-1/4	3-3/4	4-1/4	4-13/16	5-5/16
1	1.315	1-1/16	1-9/16	2-3/32	2-5/8	3-1/8	3-5/8	4-1/8	4-11/16	5-3/16
1-1/4	1.660	7/8	1-5/8	1-29/32	2-7/16	2-15/16	3-7/16	3-15/16	4-17/32	5-1/32
1-1/2	1.900	1	1-1/2	2-5/16	2-13/16	3-5/16	3-13/16	4-3/8	4-29/32	5-13/32
2	2.375	1-1/32	1-9/16	2-3/32	2-19/32	3-3/32	3-19/32	4-5/32	4-21/32	5-5/32
2-1/2	2.875	1-1/32	1-27/32	2-11/32	2-27/32	3-11/32	3-29/32	4-13/32	4-29/32	5-17/32
3	3.500	1	1-17/32	2-1/32	2-17/32	3-1/32	3-19/32	4-3/32	4-19/32	5-7/32
3-1/2	4.000	1-9/32	1-25/32	2-9/32	2-25/32	3-11/32	3-27/32	4-11/32	4-11/32	4-31/32
4	4.500	1-1/32	1-17/32	2-1/32	2-17/32	3-3/32	3-19/32	4-3/32	4-23/32	5-7/32
4-1/2	5.000	1-9/32	1-25/32	2-9/32	2-27/32	3-11/32	3-27/32	4-15/32	4-15/32	4-31/32
5	5.563	1	1-1/2	2	2-9/16	3-1/16	3-9/16	4-3/16	4-11/16	5-3/16
6	6.625	31/32	1-15/32	2-1/32	2-17/32	3-1/32	3-21/32	4-5/32	4-21/32	5-7/32
7	7.625	.....	1-17/32	2-1/32	2-17/32	3-5/32	3-21/32	4-5/32	4-21/32	5-5/32
8	8.625	.....	1-17/32	2-1/32	2-21/32	3-5/32	3-21/32	4-5/32	4-21/32	5-5/32
9	9.625	.....	1-17/32	2-5/32	2-21/32	3-5/32	3-21/32	4-5/32	4-21/32	5-5/32
10	10.750	.....	1-19/32	2-3/32	2-19/32	3-3/32	3-19/32	4-3/32	4-19/32	5-3/32
11	11.750	.....	1-19/32	2-3/32	2-19/32	3-3/32	3-19/32	4-3/32	4-19/32	5-3/32
12	12.750	.....	1-9/16	2-1/16	2-9/16	3-1/16	3-9/16	4-1/16	4-19/32	5-3/32
14	14.000	.....	1-7/16	1-15/16	2-7/16	2-15/16	3-7/16	3-15/16	4-7/16	4-15/16
Over 14 up to and including 33	.....	.....	1-7/16	1-15/16	2-7/16	2-15/16	3-7/16	3-15/16	4-7/16	4-15/16

Note. The inner diameter of pipe insulation listed in table I approximates the outer diameter of iron pipe sizes. The insulation can be applied either to a pipe or as a second layer to a smaller size of pipe insulation.

[1] Pipe insulation may be furnished in two layers to form total required thickness where the nominal thickness required is in excess of 2 inches.

3.4.3 *Class C.* Aluminum jackets shall be fabricated of 16 mil (0.016 inch) thick, 3003 alloy, having either a smooth or embossed surface. The aluminum shall be formed to fit the cylinder and have a lock seam which permits opening, closing, and locking about the cylinder. The cylinder shall have a moisture-barrier securely attached with sufficient overlap to prevent moisture migration between the insulation and the aluminum jacket. An aluminum butt strap shall be furnished with each aluminum jacket. Butt straps shall be formed in a manner which will resist penetration and degradation of the insulation by windborne water.

3.4.4 *Class D.* Kraft paper-aluminum foil jackets shall be fabricated of a kraft paper and aluminum foil lamination. The basis weight of kraft paper shall be not less than 30 pounds per 3000 square feet and aluminum foil shall be not less than 0.00035 inch thick. The water vapor transmission rate shall be not more than 0.02 perms.

3.4.4.1 *Flame spread.* Flame spread of the paper surface of the laminate shall not exceed a classification of 25.

3.4.4.2 *Puncture resistance.* Puncture resistance of the laminate shall be not less than 50 inch-ounces per inch of tear.

3.4.5 *Class E.* Dual temperature jackets shall be three ply laminates of kraft paper, foil, and kraft paper. The weight of paper, thickness of aluminum foil, and water vapor transmission rate shall be limited as stated for class D.

3.4.5.1 *Flame spread.* Flame spread of the paper surface of the laminate shall not exceed a classification of 25.

3.4.5.2 *Puncture resistance.* Puncture resistance of the laminate shall be not less than 50 inch-ounces per inch of tear.

3.4.6 *Class F.* Other jackets shall be fabricated of asphalt saturated asbestos felts, asphalt saturated and coated asbestos felts,

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polyvinyl fluoride (PVF) laminated neoprene, neoprene saturated asbestos felt, and other plastic film laminates.

3.5 Workmanship. The insulation and jackets, when furnished, shall be clean and free of dirt, grease, or other foreign matter. Corners and edges shall be sound and square. The insulation and jackets shall be free of holes. Dents, gouges, and depressions shall not be permitted in the insulation or the aluminum jackets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 *Lot.* Unless otherwise specified (see 6.2), a lot shall consist of all insulation of the same type, class, and size submitted for inspection at the same time.

TABLE II. *Classification of defects*

Defects	Major	Minor
Canvas jacketed pipe does not overlap on seam end as specified, or does not have proper adhesion to pipe (blisters or loose areas)	X	
Aluminum jacketed pipe does not have proper lock seam (loose or open or too tight for proper assembly)	X	
Aluminum jacket extends over butt ends of insulation	X	
Aluminum butt strap damaged or missing	X	
Type of insulation not as specified	X	
Dimensions not as specified or within tolerances	X	
Corners and edges not square	X	
Holes puncturing insulation	X	
Cotton jacketed pipe not free from dirt, grease, or other foreign matter		X
Dents or depressions in blocks and molded pipe insulation		X
Damage or defects not affecting function or serviceability		X
Damage or defects affecting function or serviceability	X	

4.2.2 *Visual and dimensional examination.* Sampling for visual and dimensional defects shall be in accordance with ASTM C 390. Samples shall be examined for the defects listed in table II. The AQL shall be in accordance with ASTM C 390.

4.2.3 *Sampling for tests.*

4.2.3.1 *Flexural strength and density.* Sampling and AQL for flexural strength and density tests shall be in accordance with ASTM C 390.

4.2.3.2 *Other tests.* Sampling for all other tests shall be in accordance with MIL-STD-105, level S-2. AQL shall be 6.5 percent defective.

4.2.4 *Inspection of preparation for delivery.* The packaging, packing, and marking of the block and pipe covering shall be inspected to determine conformance with applicable requirements of section 5.

4.3 Tests.

4.3.1 *Insulation.* Insulation shall be tested as shown in table III.

TABLE III. *Insulation tests*

Test	ASTM method of test	
	Type I	Type II
Density	C 303	C 302
Thermal conductivity	C 177	C 335
Compressive strength	C 165	.....
Flexural strength	C 203	C 446
Linear shrinkage	C 356	C 356

4.3.2 *Jackets.* Kraft paper-aluminum foil jackets (class D) and dual temperature jackets (class E) shall be tested as shown in table IV.

TABLE IV. *Jacket tests*

Test	ASTM method of test
Water vapor transmission	E 96, method A
Flame spread	E 84
Puncture resistance	D 781

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

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5.1.1 *Level A.* Blocks or pipe covering of like description shall be packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed and waterproof sealed in accordance with the appendix to the box specification.

5.1.2 *Level B.* Blocks or pipe covering of like description shall be packaged in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 *Level C.* Blocks or pipe covering shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Blocks or pipe covering of like description shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, overseas type, type III; PPP-B-621, class 2; PPP-B-636, class weather resistant; or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple-wall fiberboard box and the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification. Unit packages of blocks or pipe covering of like description may be packed in a crate conforming to PPP-B-650 or on a pallet conforming to MIL-STD-147. Fiberboard boxes shall be waterproof sealed with tape in accordance with the applicable box specification and appendix thereto. Wood boxes shall be provided with case liners conforming to MIL-L-10547 sealed in accordance with the appendix thereto. Crates shall be provided with shrouds in accordance with the appendix to PPP-B-650.

5.2.2 *Level B.* Blocks or pipe covering of like description shall be packed in accordance with 5.2.1, except the boxes shall be the domestic type or class.

5.2.3 *Level C.* Blocks or pipe covering shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates, in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to markings required by the contract or purchase order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military activities.* In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Block and pipe insulation is intended for insulating surfaces up to 1200°F.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type and class as applicable for pipe insulation (1.2.1).
(*c*) Other jacket material required when class F is specified (see 3.4.6).
(*d*) Shape and size of block required (see 3.2).
(*e*) Size and thickness of pipe insulation (see 3.3).
(*f*) Size of lot, if different from 4.2.1.
(*g*) Selection of applicable levels of packaging and packing required (see 5.1, 5.2, and 6.3).

6.3 Level B packaging and packing is intended to provide economical but limited protection and should be specified only when it is determined the insulation will be held in covered storage.

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